Nama : Evelyn Anastasia

NIM : 2312101

Kelas : 2 B

UJIAN AKHIR SEMESTER - ALGORITMA & PEMOGRAMAN

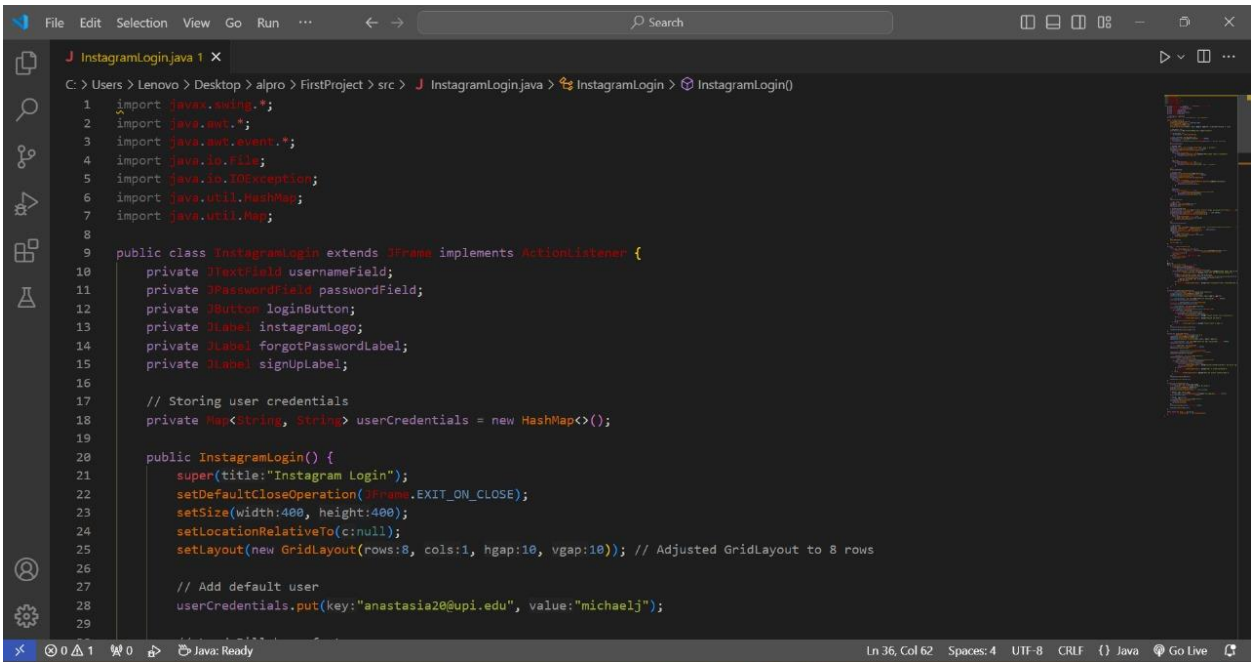
LOGIN INSTAGRAM

GUI (Graphical User Interface) sebuah aplikasi adalah antarmuka yang memungkinkan pengguna untuk berinteraksi dengan perangkat lunak atau sistem komputer melalui elemen-elemen grafis seperti ikon, tombol, menu, jendela. GUI mempermudah Penggunaan perangkat karena lebih intuitif dan visual. Jadi dengan adanya GUI, kita dapat mengetahui bahwa apa yang kita input telah diterima dan outputnya ditunjukkan secara visual.

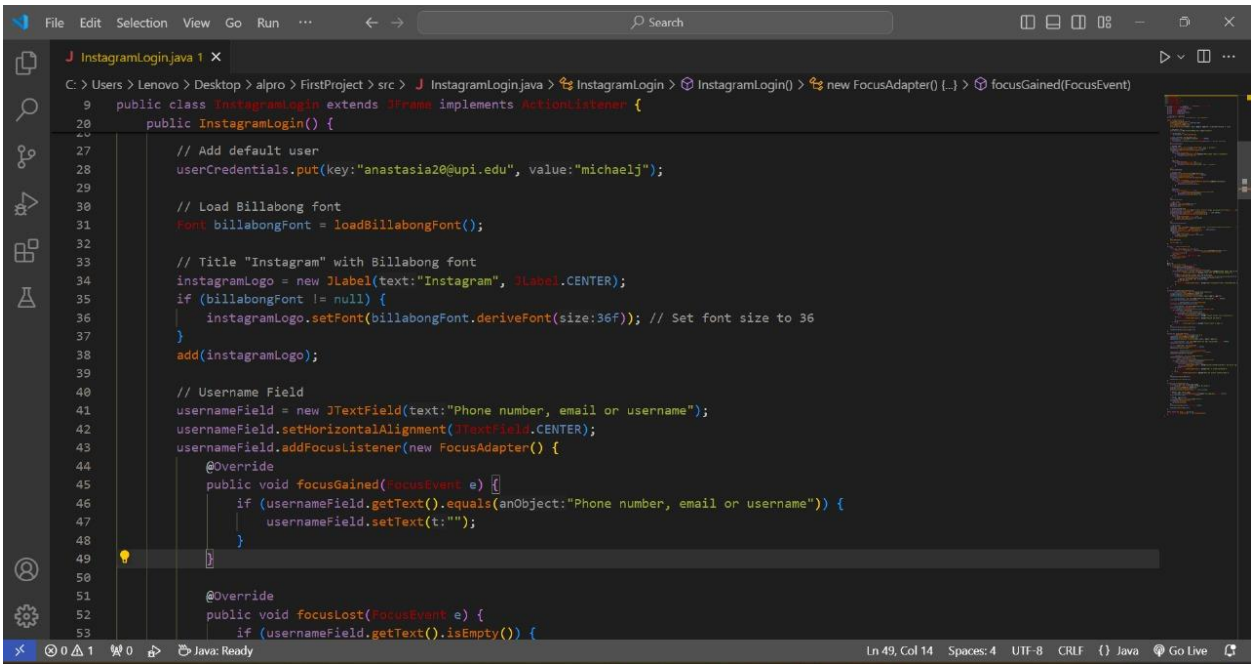
Mengimplementasikan Login Instagram Menggunakan GUI Java

Program ini mengimplementasikan *form login* Instagram. Langkah-langkah:

1. Masuukkan email dan *password*, kemudian tekan *button login*, makan akan muncul *pop up* “anda berhasil *login*” lalu akan dialihkan pada halaman *main window* yang seharusnya berisi gambar serta terdapat *button logout* untuk mengeluarkan akun dan akan Kembali ke halaman *login*.
2. Jika lupa *password*, anda dapat mengubah *password* diganti dengan yang baru, lalu *login* kembali.
3. Jika tidak memiliki akun, buatlah akun dengan cara menekan *button* “Don’t have an account? Sign up.”, lalu masukkin email dan *password* untuk membuat akun abru. Setelah berhasil membuat akun baru, maka akan muncul *pop out* “Account created successfully. You can now login.”

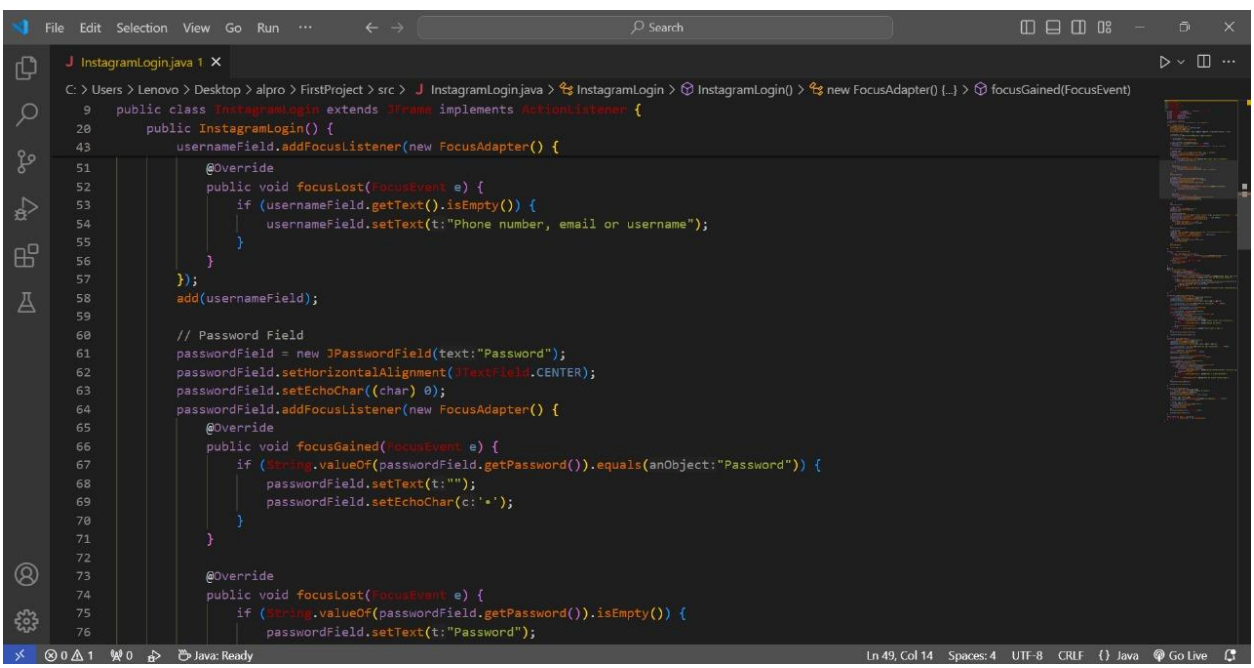


Gambar 1.1



```java
public class InstagramLogin extends JFrame implements ActionListener {
    public InstagramLogin() {
        // Add default user
        userCredentials.put(key:"anastasia20@upi.edu", value:"michaelj");

        // Load Billabong font
        Font billabongFont = loadBillabongFont();

        // Title "Instagram" with Billabong font
        instagramLogo = new JLabel(text:"Instagram", JLabel.CENTER);
        if (billabongFont != null) {
            instagramLogo.setFont(billabongFont.deriveFont(size:36f)); // Set font size to 36
        }
        add(instagramLogo);

        // Username Field
        usernameField = new JTextField(text:"Phone number, email or username");
        usernameField.setHorizontalAlignment(JTextField.CENTER);
        usernameField.addFocusListener(new FocusAdapter() {
            @Override
            public void focusGained(FocusEvent e) {
                if (usernameField.getText().equals(anObject:"Phone number, email or username")) {
                    usernameField.setText(t:"");
                }
            }

            @Override
            public void focusLost(FocusEvent e) {
                if (usernameField.getText().isEmpty()) {
```

Gambar 1.2



```java
public class InstagramLogin extends JFrame implements ActionListener {
    public InstagramLogin() {
        usernameField.addFocusListener(new FocusAdapter() {
            @Override
            public void focusLost(FocusEvent e) {
                if (usernameField.getText().isEmpty()) {
                    usernameField.setText(t:"Phone number, email or username");
                }
            }
        });
        add(usernameField);

        // Password Field
        passwordField = new JPasswordField(text:"Password");
        passwordField.setHorizontalAlignment(JTextField.CENTER);
        passwordField.setEchoChar((char) 0);
        passwordField.addFocusListener(new FocusAdapter() {
            @Override
            public void focusGained(FocusEvent e) {
                if (String.valueOf(passwordField.getPassword()).equals(anObject:"Password")) {
                    passwordField.setText(t:"");
                    passwordField.setEchoChar(c:'•');
                }
            }

            @Override
            public void focusLost(FocusEvent e) {
                if (String.valueOf(passwordField.getPassword()).isEmpty()) {
                    passwordField.setText(t:"Password");
```

Gambar 1.3

```java
public class InstagramLogin extends JFrame implements ActionListener {
    public InstagramLogin() {
        passwordField.addFocusListener(new FocusAdapter() {
            public void focusLost(FocusEvent e) {
                if (String.valueOf(passwordField.getPassword()).isEmpty()) {
                    passwordField.setText(t:"Password");
                    passwordField.setEchoChar((char) 0);
                }
            }
        });
        add(passwordField);

        // Login Button
        loginButton = new JButton(text:"Log In");
        loginButton.addActionListener(this);
        add(loginButton);

        // Forgot Password Label
        forgotPasswordLabel = new JLabel(text:"<html><a href=\"#\">Forgot your password?</a></html>", JLabel.CENTER);
        forgotPasswordLabel.setForeground(Color.BLUE.darker());
        forgotPasswordLabel.setCursor(Cursor.getPredefinedCursor(Cursor.HAND_CURSOR));
        forgotPasswordLabel.addMouseListener(new MouseAdapter() {
            @Override
            public void mouseClicked(MouseEvent e) {
                // Open a new window to reset the password
                openResetPasswordWindow();
            }
        });
        add(forgotPasswordLabel);
```

Gambar 1.4

```java
public class InstagramLogin extends JFrame implements ActionListener {
    public InstagramLogin() {
        // Sign Up Label
        signUpLabel = new JLabel(text:"<html><a href=\"#\">Don't have an account? Sign up.</a></html>", JLabel.CENTER);
        signUpLabel.setForeground(Color.BLUE.darker());
        signUpLabel.setCursor(Cursor.getPredefinedCursor(Cursor.HAND_CURSOR));
        signUpLabel.addMouseListener(new MouseAdapter() {
            @Override
            public void mouseClicked(MouseEvent e) {
                // Open a new window to create a new account
                openSignUpWindow();
            }
        });
        add(signUpLabel);

        setVisible(b:true);
    }

    private Font loadBillabongFont() {
        try {
            // Load the Billabong font from the file
            Font font = Font.createFont(Font.TRUETYPE_FONT, new File(pathname:"Billabong.ttf"));
            GraphicsEnvironment ge = GraphicsEnvironment.getLocalGraphicsEnvironment();
            ge.registerFont(font);
            return font;
        } catch (FontFormatException | IOException e) {
            e.printStackTrace();
            return null;
        }
```

Gambar 1.5

```java
.ic class InstagramLogin extends JFrame implements ActionListener {
private Font loadBillabongFont() {
    }
}

@Override
public void actionPerformed(ActionEvent e) {
    if (e.getSource() == loginButton) {
        // Validate username and password
        String username = usernameField.getText();
        String password = new String(passwordField.getPassword());
        if (username.isEmpty() || password.isEmpty() || username.equals(anObject:"Phone number, email or username") || password.equals(anOb
            JOptionPane.showMessageDialog(this, message:"Please enter your username and password.");
        } else {
            // Check if username and password match the stored values
            if (userCredentials.containsKey(username) && userCredentials.get(username).equals(password)) {
                JOptionPane.showMessageDialog(this, message:"Anda berhasil login.");
                // Open the main window after successful login
                openMainWindow();
            } else {
                JOptionPane.showMessageDialog(this, message:"Email atau password salah. Silakan buat akun baru.");
            }
        }
    }
}

private void openResetPasswordWindow() {
    JFrame resetPasswordWindow = new JFrame(title:"Reset Password");
    resetPasswordWindow.setSize(width:300, height:150);
```

Gambar 1.6

```java
.ic class InstagramLogin extends JFrame implements ActionListener {
private void openResetPasswordWindow() {
    JFrame resetPasswordWindow = new JFrame(title:"Reset Password");
    resetPasswordWindow.setSize(width:300, height:150);
    resetPasswordWindow.setLocationRelativeTo(c:null);
    resetPasswordWindow.setLayout(new GridLayout(rows:3, cols:1, hgap:10, vgap:10));

    JLabel instructionLabel = new JLabel(text:"Enter your new password:", JLabel.CENTER);
    resetPasswordWindow.add(instructionLabel);

    JPasswordField newPasswordField = new JPasswordField();
    resetPasswordWindow.add(newPasswordField);

    JButton confirmButton = new JButton(text:"Confirm");
    confirmButton.addActionListener(e -> {
        String newPassword = new String(newPasswordField.getPassword());
        if (!newPassword.isEmpty()) {
            String username = usernameField.getText();
            if (userCredentials.containsKey(username)) {
                userCredentials.put(username, newPassword);
                JOptionPane.showMessageDialog(this, message:"Password has been reset successfully.");
            } else {
                JOptionPane.showMessageDialog(this, message:"Username not found.");
            }
            resetPasswordWindow.dispose();
        } else {
            JOptionPane.showMessageDialog(this, message:"Password cannot be empty.");
        }
    });
```

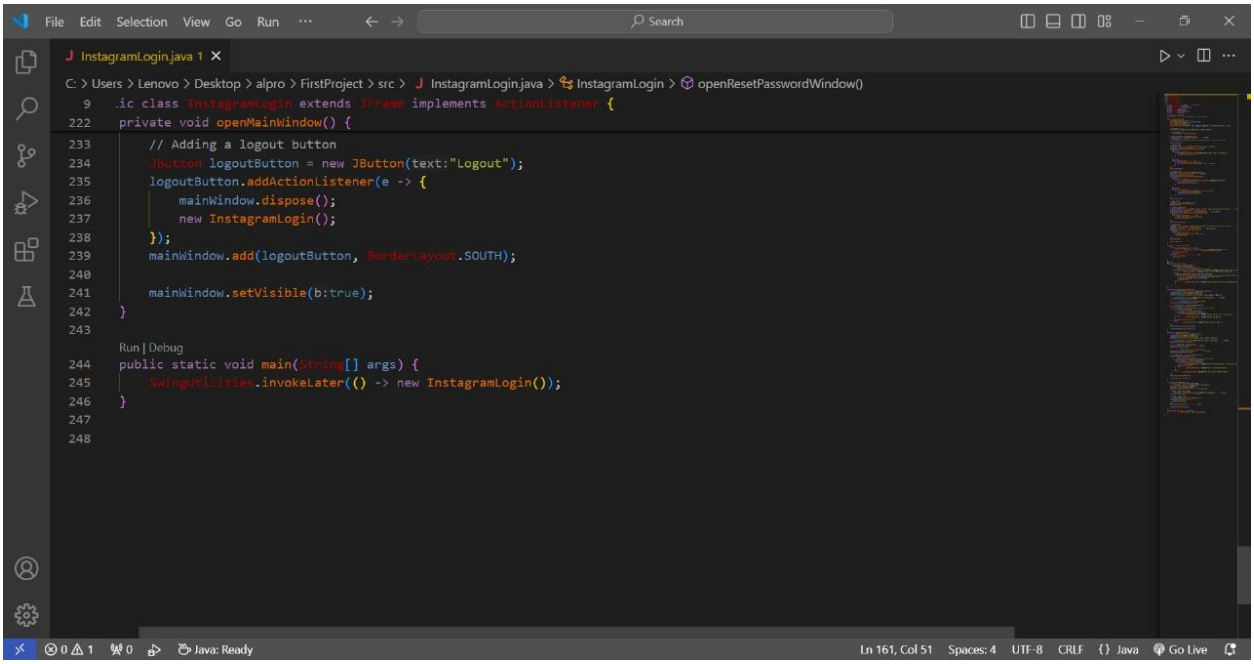
Gambar 1.7

```java
.ic class InstagramLogin extends JFrame implements ActionListener {
private void openResetPasswordWindow() {
    resetPasswordWindow.add(confirmButton);

    resetPasswordWindow.setVisible(b:true);
}

private void openSignUpWindow() {
    JFrame signUpWindow = new JFrame(title:"Sign Up");
    signUpWindow.setSize(width:300, height:200);
    signUpWindow.setLocationRelativeTo(c:null);
    signUpWindow.setLayout(new GridLayout(rows:4, cols:1, hgap:10, vgap:10));

    JLabel instructionLabel = new JLabel(text:"Enter your email and password:", JLabel.CENTER);
    signUpWindow.add(instructionLabel);

    JTextField emailField = new JTextField();
    emailField.setHorizontalAlignment(JTextField.CENTER);
    signUpWindow.add(emailField);

    JPasswordField newPasswordField = new JPasswordField();
    newPasswordField.setHorizontalAlignment(JPasswordField.CENTER);
    signUpWindow.add(newPasswordField);

    JButton signUpButton = new JButton(text:"Sign Up");
    signUpButton.addActionListener(e -> {
        String email = emailField.getText();
        String newPassword = new String(newPasswordField.getPassword());
        if (!email.isEmpty() && !newPassword.isEmpty()) {
```

Gambar 1.8

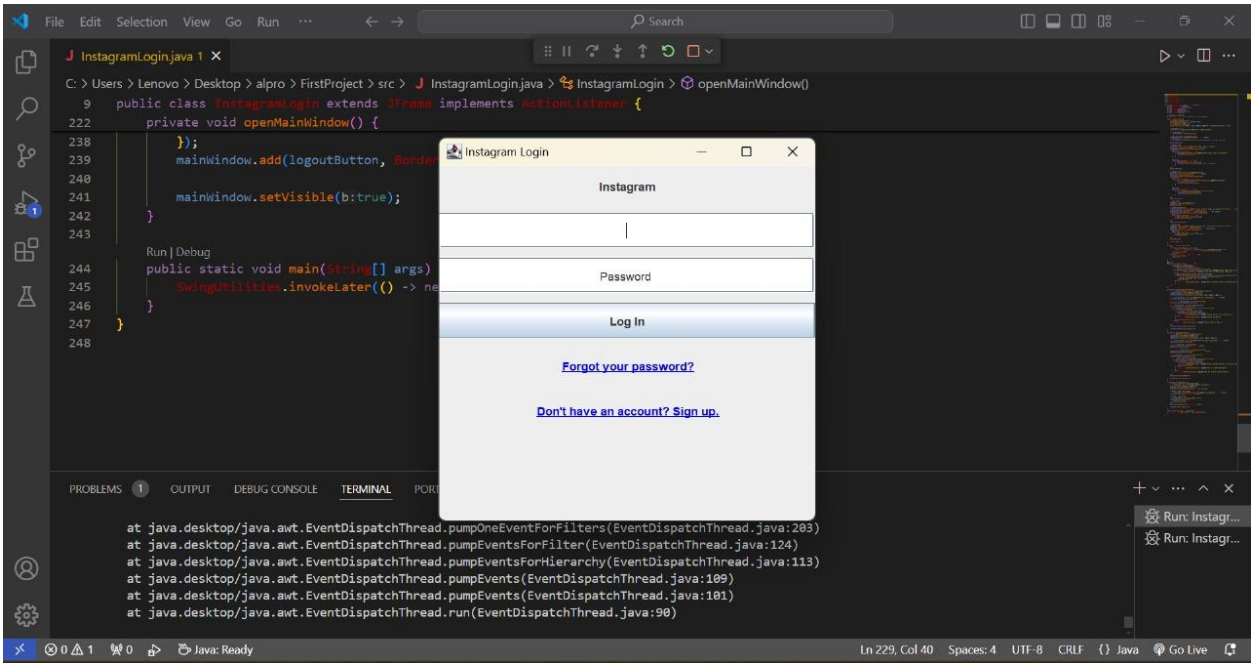
```java
.ic class InstagramLogin extends JFrame implements ActionListener {
private void openSignUpWindow() {
                userCredentials.put(email, newPassword);
                JOptionPane.showMessageDialog(this, message:"Account created successfully. You can now log in.");
                signUpWindow.dispose();
            } else {
                JOptionPane.showMessageDialog(this, message:"Email is already registered.");
            }
        } else {
            JOptionPane.showMessageDialog(this, message:"Email and password cannot be empty.");
        }
    });
    signUpWindow.add(signUpButton);

    signUpWindow.setVisible(b:true);
}

private void openMainWindow() {
    dispose(); // Close the current login window
    JFrame mainWindow = new JFrame(title:"Instagram Main Window");
    mainWindow.setSize(width:600, height:600);
    mainWindow.setLocationRelativeTo(c:null);
    mainWindow.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

    // Adding a label with an image
    JLabel mainImage = new JLabel(new ImageIcon(filename:"main_image.png"), JLabel.CENTER);
    mainWindow.add(mainImage, BorderLayout.CENTER);

    // Adding a logout button
```

gambar 1.9



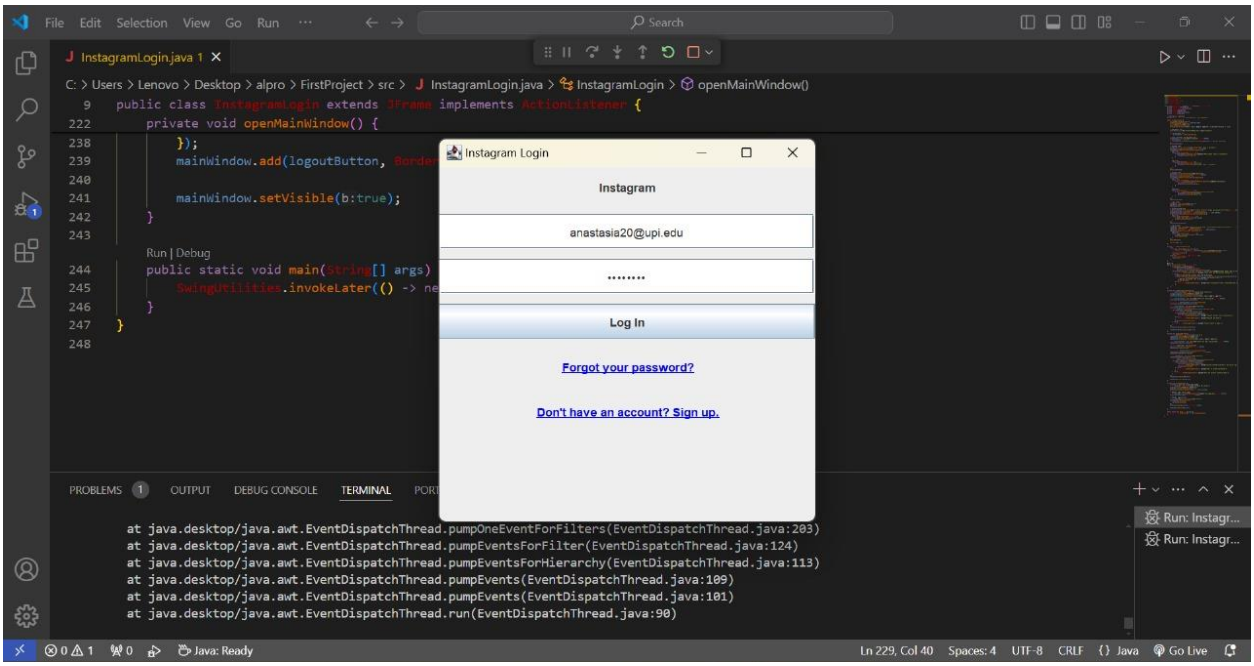
Gambar 1.10

Outputnya akan menghasilkan seperti ini.



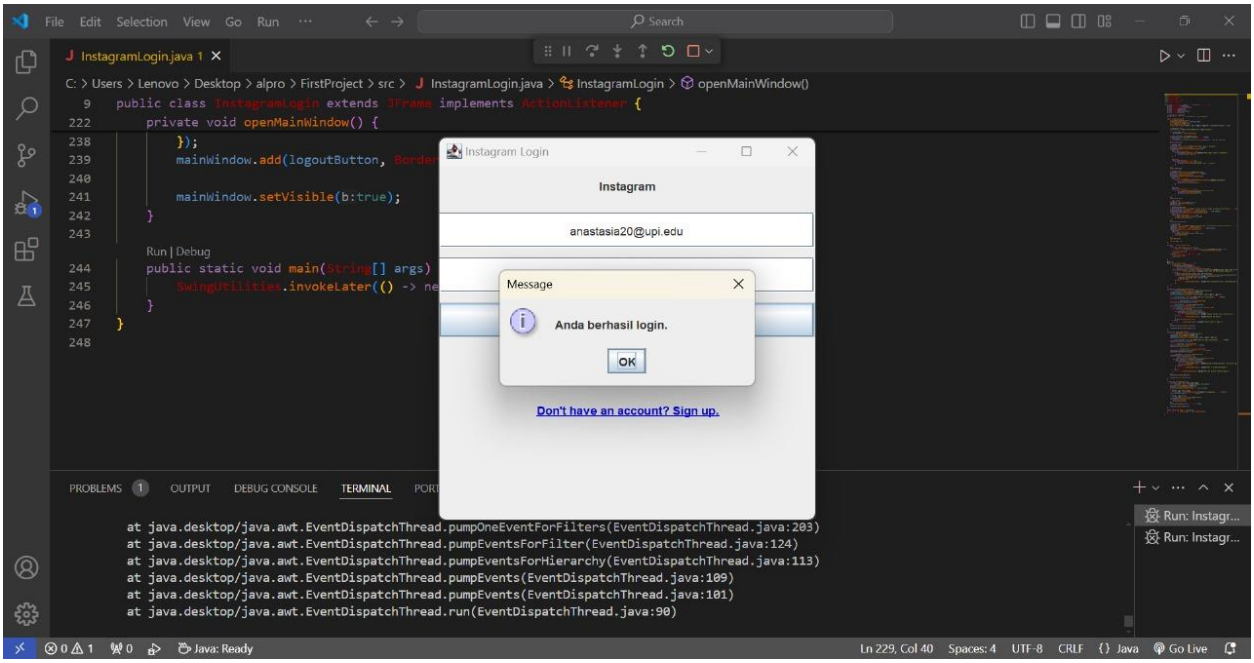
Gambar 2.1

Jika anda sudah memiliki akun, anda dapat masukkan alamat email dan password, lalu tap button login.



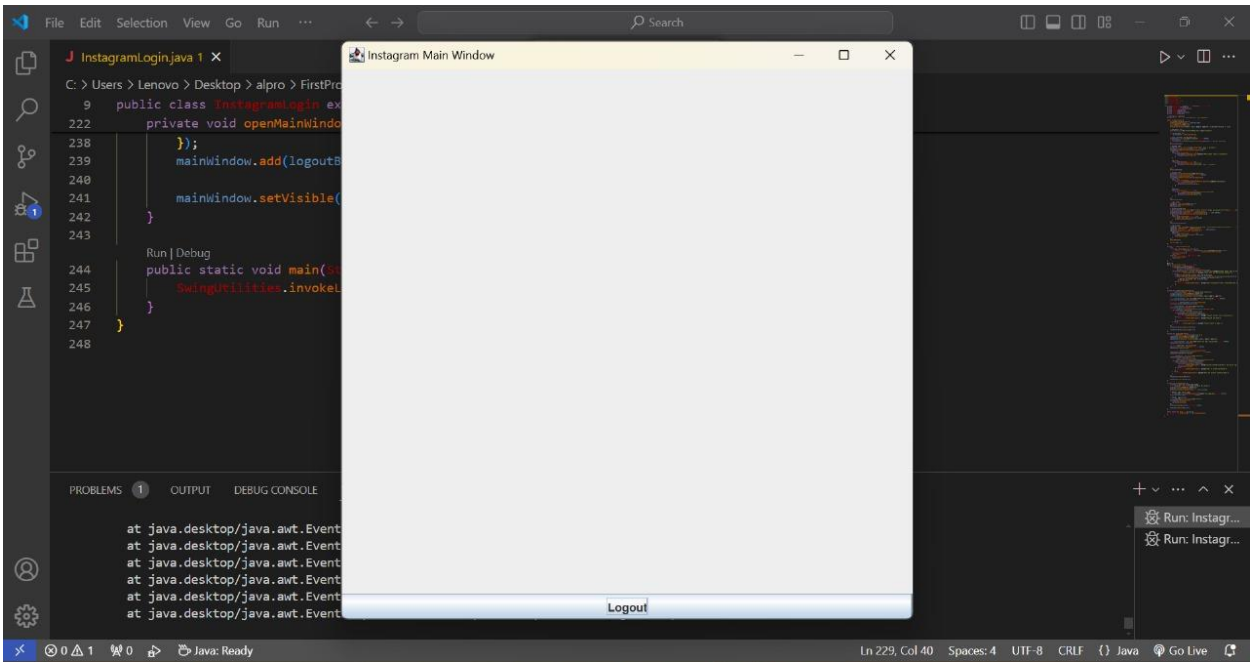
gambar 2.2

Maka, anda berhasil login.



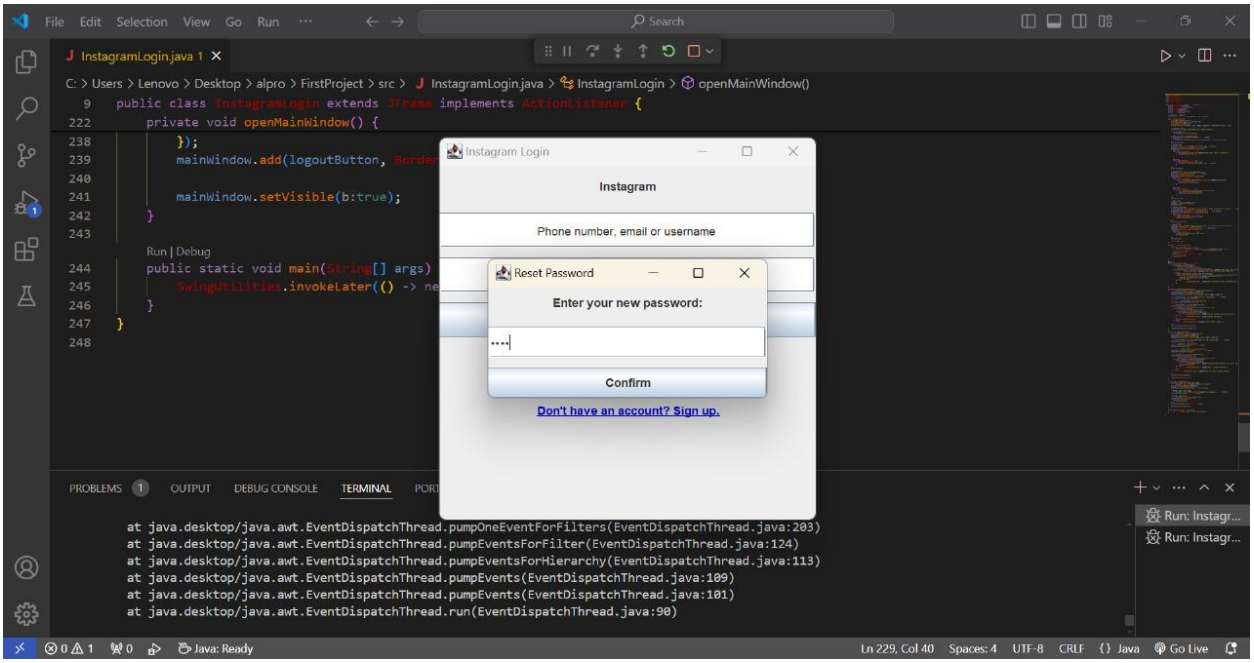
Gambar 2.3

Setelah anda login, anda akan dialihkan pada halaman “main window” yang seharusnya ada foto, dan dibawahnya terdapat button log out.



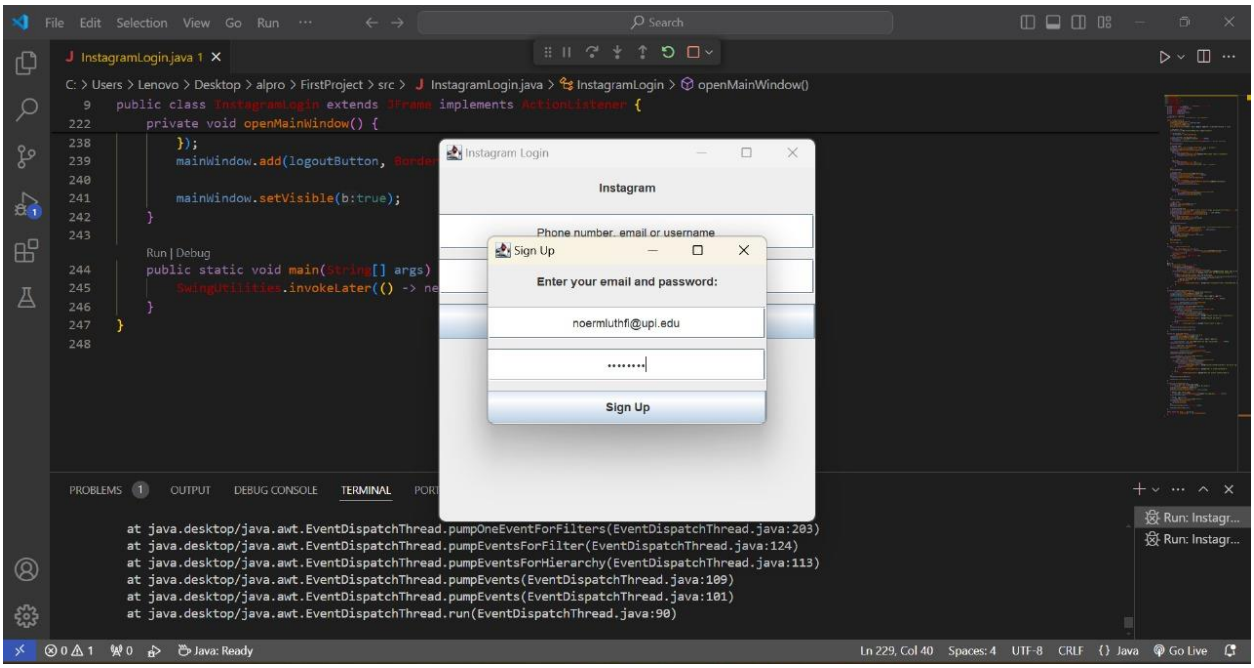
Gambar 2.4

Jika anda lupa password, anda dapat mengubah password.



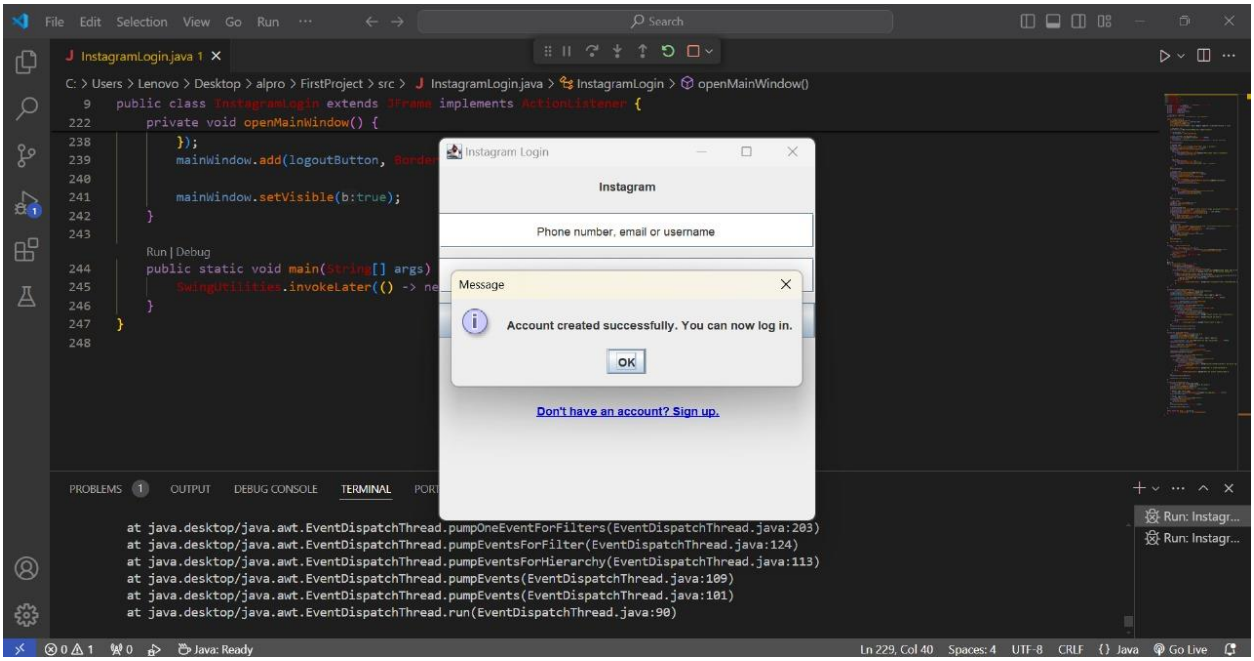
Gambar 2.5

Anda juga dapat menambahkan akun baru dengan memasukkan email dan password, lalu klik “sign up”.



Gambar 2.6

Jika anda sudah membuat akun, maka akan muncul *pop up* yang mengeluarkan halaman “account successfully, you are now login.”



Gambar 2.7